

# Reliable AI Writing Systems

## 1. Problem Decomposition: The Anatomy of Hallucinations in Long-Form Legal Writing

Hallucinations in 5–20 page judicial documents are not a single technical failure but a multivariable phenomenon. In a legal context where facts are the primary "currency," a hallucination is not just an accuracy error; it is a direct threat to due process and judicial neutrality.

### A. Taxonomy of Error in the Judicial Domain

To solve the problem, we must distinguish between failure types based on their legal impact:

- **Reference Hallucinations (Citations & Authorities):** The invention or blending of case law, docket numbers, or statutes. This is the most critical error, as it induces a judge to base a ruling on non-existent legal premises.
- **Factual Hallucinations (Fact-Morphing):** Slight alterations of factual details (e.g., changing "the witness was unsure" to "the witness saw nothing"). In law, qualifiers and adverbs define the application of a rule; a minimal change constitutes a substantive hallucination.
- **Global Coherence Contradictions (Long-Context Drifting):** In lengthy documents, models may maintain local coherence while failing globally—affirming a premise on page 2 and contradicting it on page 18 due to context window limits or "lost-in-the-middle" issues.
- **Hallucinations by Selective Omission:** Not an explicit lie, but the exclusion of exculpatory facts or key evidence present in the record. This creates a "hallucinated narrative" that is logically sound but factually incomplete.

### B. Vulnerabilities in Pro-Se Litigation

A unique challenge for judicial systems is analyzing cases where parties represent themselves (*pro-se*).

- **Over-structuring Informal Narratives:** Citizens often present facts in an emotional or disorganized manner. A model, in an attempt to be "helpful," may hallucinate a "legal intent" never expressed by the citizen, mis-translating informal complaints into incorrect technical legal figures.
- **Helpfulness Bias:** Under Reinforcement Learning from Human Feedback (RLHF) training, models tend to be compliant. When faced with informative gaps in a *pro-se* account, the AI may "fill in the blanks" to make the argument sound stronger, breaking the impartiality required by the court.

### C. Root Causes

- **Anaphoric Ambiguity:** As context windows fill, the model's ability to resolve subjects (e.g., "the aforementioned") degrades, leading to "hallucinated attribute assignment" where actions are attributed to the wrong party.

- **Probability vs. Truth:** Models are optimized for verisimilitude (sounding correct), not veracity. Legal verisimilitude is easy to fake, but veracity is binary.
  - **Lossy Compression:** When summarizing long records, models abstract information. Legal nuances—the essence of a judgment—are often discarded as "noise" and replaced by hallucinated generalizations.
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## 2. Evaluation Rubric: Defining "Solved" for Judicial Systems

We propose an evaluation framework based on **Atomic Fidelity** and **Contradiction Management**.

### A. Primary Metric: Verified Claim Precision (VCP)

The system decomposes output into **Atomic Information Units (AIU)**—the smallest expression of a fact or citation that can be independently verified.

$$\text{VCP} = \text{Verified AIUs with Evidence} / \text{Total Generated AIUs}$$

**Ground Truth Challenge:** Success is measured against "**The Record**". If the record contains a lie, the system is "successful" if it reports that lie accurately, not if it tries to correct it.

### B. Practical AIU Evaluation Example

- **Generated Sentence:** "The defendant, Juan Pérez, delivered the notice on March 14, 2024."
  - **AIU-1 (Subject/Role):** Is Juan Pérez the defendant? (**Check**)
  - **AIU-2 (Action):** Was it a "delivery" and not just an "advice"? (**Check**)
  - **AIU-3 (Temporal):** Was it March 14? (**Fail: Record states March 15**)
- **Result:** A single false AIU invalidates the entire sentence (0% precision), as a date error can trigger statute of limitations issues.

### C. Systemic Trade-offs

- **Precision over Recall:** We prefer the AI to omit a fact (Low Recall) than to invent one (Low Precision). A system that admits "I cannot verify this data" maintains legitimacy.
  - **Accuracy over Cost:** The cost of a judicial hallucination (mistrial) is infinite compared to computational costs. We prioritize "Slow Reasoning" models and multi-agent loops.
  - **Reliability over Latency:** We accept 3–5 minutes of processing for 20 pages to allow for background Reflection and Fact-Checking cycles.
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### 3. Proposed Architecture: The Adversarial Auditor-Editor Loop

Our architecture rejects the standard "linear pipeline" in favor of a **Zero-Trust, multi-agent framework** designed to enforce factual grounding at every stage of the generation process.

#### A. Phase I: Deterministic Grounding & Entity Anchoring

The process begins with **Deterministic Ingestion**. Rather than feeding a raw 20-page PDF into a model, we perform **Entity-Centric Chunking**. This stage builds a persistent **Entity Map** that tracks roles (Plaintiff, Defendant, Witness) and their associated metadata across the entire document. This prevents "Lost-in-the-Middle" subject confusion, ensuring a consistent "source of truth" for all parties.

#### B. Phase II: The Adversarial "Auditor-Editor" Cycle

The core of the proposal is the separation of powers between two specialized agents with conflicting incentives:

- **The Writer Agent:** Optimized for high-level synthesis, global coherence, and judicial tone. It generates draft units in AIUs.
- **The Auditor Agent:** This agent is explicitly prompted to **invalidate** the Writer. For every AIU produced, the Auditor suspends generation and executes a **Precision RAG query** against the record.
  - **Self-Correction Loop:** If a discrepancy is found, it triggers a **Reflection & Correction** step, forcing a rewrite until the claim matches the evidence.
  - **Conflict-Aware Arbitrage:** If the Auditor finds genuine factual friction (e.g., Witness A contradicts Witness B), it triggers a **Human Clarification Request**. This ensures the judge—not the model—resolves factual tension.

#### C. Phase III: The Deterministic "Citations Firewall"

To eliminate Level 1 hallucinations, we implement a **non-probabilistic firewall** for legal authorities. Any legal reference is intercepted and cross-referenced via API against verified external legal databases (e.g., LexisNexis). If the citation is unverified or a jurisdictional mismatch occurs, it is blocked.

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### 4. Validation Plan: Reliability and Intellectual Honesty

#### A. Two-Week Execution

- **Week 1: Adversarial Red Teaming:** We will insert "contaminated facts" into 100 real records (e.g., shifting dates by one day). **Success Metric:** 100% Recall in detecting induced hallucinations.
- **Week 2: Human Trust & Alert Fatigue:** We will deliver reports to 20 clerks and measure the **"Fatigue Metric"**—how many flags are required before a user begins to ignore them (Automation Bias).

## B. Open Questions & Assumptions

- **Ingestion Poisoning:** We assume the initial **Entity Map** is correct. If the AI confuses the plaintiff and defendant during pre-processing, all subsequent audits will validate facts against a flawed schema.
- **RAG Confirmation Bias:** If the retrieval system fails to find the key paragraph, the Auditor will lack the evidence to refute a Writer's hallucination.

## C. Kill Switches

We will abandon this approach if:

- Verification costs equal the time it takes a clerk to perform the task manually.
- The system generates a **Level 1 False Negative** (fatal citation hallucination) in the adversarial dataset.